n-types are closed under retracts: a retract of an n-type is an n-type

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'thm:h-level-retracts' (source lines 68-73, Theorem 7.1.4); the proof environment begins after one blank line, so the pairing is by adjacency.

Editorial note. The retained context is the author's own lead-in sentence, which states that the theory of n-types begins with closure under operations and constructors.

Editorial note. The retained counter values (chapter 7, section 1, theorem offset 3) make the retained theorem print as Theorem 7.1.4, matching the original book numbering.

Editorial note. The two external references point at results of earlier chapters that the retained proof invokes by label; they are navigation pointers only.

Editorial scope. One original result of the HoTT Book, counted once: closure of n-types under retracts (Theorem 7.1.4).

Allowed standard prerequisites (editorial).

- n-types. X is an n-type if its n-fold iterated path spaces are contractible, with the cases n = -2 (contractible) and n = -1 (mere proposition) as the base of the hierarchy.. The proof is an induction on n whose base case is contractibility and whose step reduces the claim to a statement about path spaces.
- Retractions. p : X -> Y is a retraction if it has a section s with a homotopy p o s ~ 1.. The section s and the homotopy epsilon are the data used to build the retraction of path spaces at source lines 81-87.
- Retracts of contractible types are contractible. A retract of a contractible type is contractible.. It is the base case n = -2 at source line 77.

Source reference. Original theorem 3.11.7: sources/hott-hlevels/book-labelnumbers-first-edition.tex, source lines 1875–1875. Original result cited by the retained text but located outside the retained span. It is a navigation pointer: the counted statement's own author proof is retained line for line, and no part of it is replaced by this reference.

Source reference. Original lemma 2.4.3: sources/hott-hlevels/book-labelnumbers-first-edition.tex, source lines 1540–1540. Original result cited by the retained

text but located outside the retained span. It is a navigation pointer: the counted statement's own author proof is retained line for line, and no part of it is replaced by this reference.

We begin the general theory of n-types by showing they are closed under certain operations and constructors.

Theorem 7.1.4. *Let $p : X \to Y$ be a retraction and suppose that X is an n-type, for any $n \geq -2$. Then Y is also an n-type.*

Proof. We proceed by induction on n. The base case $n = -2$ is handled by original theorem 3.11.7.

For the inductive step, assume that any retract of an n-type is an n-type, and that X is an $(n+1)$-type. Let $y, y' : Y$; we must show that $y = y'$ is an n-type. Let s be a section of p, and let ϵ be a homotopy $\epsilon : p \circ s \sim 1$. Since X is an $(n+1)$-type, $s(y) =_X s(y')$ is an n-type. We claim that $y = y'$ is a retract of $s(y) =_X s(y')$. For the section, we take

$$\mathsf{ap}_s : (y = y') \to (s(y) = s(y')).$$

For the retraction, we define $t : (s(y) = s(y')) \to (y = y')$ by

$$t(q) :\equiv {\epsilon_y}^{-1} \cdot p(q) \cdot \epsilon_{y'}.$$

To show that t is a retraction of ap_s, we must show that

$${\epsilon_y}^{-1} \cdot p(s(r)) \cdot \epsilon_{y'} = r$$

for any $r : y = y'$. But this follows from original lemma 2.4.3. □